TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	87.1 ± 1.14
Correlation Stability	93.5 ± 0.50
Distribution Consistency	92.7 ± 1.90
Entropy Preservation	53.3 ± 0.46
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	98.3 ± 1.10
Privacy Risk	2.5 ± 1.28
Data Density	100.0 ± 0.00
Coverage Score	69.7 ± 4.94
Feature Accuracy	90.9 ± 2.74
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	44.661	-0.085	0.2%	OK
TB	3.299	3.765	+0.466	14.1%	Warn
DB	1.486	1.717	+0.231	15.5%	Warn
Alkphos	290.576	341.268	+50.692	17.4%	Warn
Sgpt	80.714	95.172	+14.458	17.9%	Warn
Sgot	109.911	126.043	+16.132	14.7%	Warn
TP	6.483	6.413	-0.070	1.1%	OK
ALB	3.142	3.118	-0.024	0.8%	OK
A/G Ratio	0.947	0.965	+0.018	1.9%	OK
Selector	1.286	1.290	+0.003	0.3%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	87.1 ± 1.14
Correlation Stability (*2)	93.5 ± 0.50
Distribution Consistency (*3)	92.7 ± 1.90
Entropy Preservation (*4)	53.3 ± 0.46
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	98.3 ± 1.10
Privacy Risk (*7)	2.5 ± 1.28
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	69.7 ± 4.94

Feature Accuracy (*10)	90.9 ± 2.74
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.7051282051282051	0.7042735042735042	-0.00170940170940171	
F1	0.8128002697439015	0.8194242885769576	0.8187912020170787	0.00599093227317721	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6547008547008548	0.6547008547008547	0.6735042735042736	0.01880341880341882	
F1	0.7599945800761612	0.7608643449747922	0.7758571994252744	0.01586261934911325	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.482051282051282	0.5017094017094017	-0.05726495726495726	*
F1	0.5587432119380917	0.4296617171611039 6	0.4646744129593528 6	-0.0940687989787388	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6683760683760684	0.6974358974358974	-0.00512820512820513 25	
F1	0.7997903571549948	0.7733148975206192	0.7945400039456241	-0.00525035320937070 6	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.7068376068376068	0.7068376068376068	0	
F1	0.8279091624529537	0.8279091624529537	0.8279091624529537	0	

Model: RF

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376069	0.6991452991452991	0.6957264957264957	-0.011111111111111118	
F1	0.8107962291699315	0.8071556099294893	0.8059781155946935	-0.004818113575238181	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6820512820512821	0.6846153846153846	-0.0017094017094017144	
F1	0.7971299319768493	0.7924789189803157	0.7948671089476391	-0.0022628230292102815	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6820512820512821	0.6914529914529914	0.01709401709401709	
F1	0.7821343746683586	0.790484005375441	0.7950572099012005	0.012922835232841856	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6820512820512821	0.6854700854700855	0.6982905982905983	0.016239316239316248	
F1	0.7882128262729017	0.7901767208482808	0.7984722907701687	0.010259464497267057	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6965811965811965	0.7068376068376068	0.6965811965811965	0	
F1	0.8042647469349659	0.8045508674066489	0.7995079242243317	-0.004756822210634048	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	51.0	Male	1.8656660132577816	0.6644803870354115	451.0	129.0	279.0	6.801966040191674	4.155998763871402	1.5579422128263338
2	54.0	Female	1.0121747247565542	0.2285670461991924	193.0	24.0	30.0	6.473359212325868	3.595102674309829	1.0902669325103056
3	55.0	Male	0.7970263541585519	0.2419988818022854 3	158.0	97.0	14.0	6.350869760644904	3.332779521999389	1.1149247896802516
4	73.0	Male	5.08126578335105	-3.6639876397864155	290.0	-3.0	24.0	6.815834903832169	2.159719292483758	0.4540848834335 4614
5	77.0	Female	1.0936830930870027	-0.0564951572893539	218.0	19.0	-0.0	8.014454242982874	4.14499687709676	1.1460082160972236
6	45.0	Male	1.6494576314243123	0.7712128519465944	142.0	21.0	15.0	7.020735180829028	4.634831779971435	1.4547556262908843
7	33.0	Male	7.1	3.7	196.0	622.0	497.0	6.9	3.6	1.09
8	38.0	Male	2.2	1.0	310.0	119.0	42.0	7.9	4.1	1.0
9	64.0	Male	3.0	1.4	248.0	46.0	40.0	6.5	3.2	0.9
10	40.0	Female	0.9	0.2	285.0	32.0	27.0	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	8.0	Male	1.1484675213029072	0.1435332734462 2505	79.0	49.0	43.0	8.03527676391906 8	3.92176409914167 9	1.2409644191810 91
2	33.0	Female	1.1096679317465343	0.1803759523021 4348	712.0	101.0	-22.0	7.413729690290971	4.770110755547395	1.3885459569608 474
3	43.0	Male	0.0506436368139 2714	3.428034755582934	163.0	27.0	83.0	6.510571808859118	5.224509619966549	1.2728962805082 902
4	49.0	Male	0.7729508434144 92	0.1472401249516 637	246.0	23.0	38.0	5.845786356250249	3.789937583982613	1.1014178059721 08
5	51.0	Male	1.2355043637940948	0.9498209068445 385	809.0	94.0	128.0	5.464946008682229 5	4.004485398390199	1.0283092242999 48
6	61.0	Male	14.081455357675829	0.4882263604407 4574	481.0	43.0	60.0	6.343817204527677	3.179472641224817	0.8637348472926 022
7	50.0	Female	1.0	0.3	191.0	22.0	31.0	7.8	4.0	1.0
8	52.0	Female	0.6	0.1	194.0	10.0	12.0	6.9	3.3	0.9
9	30.0	Male	0.8	0.2	182.0	46.0	57.0	7.8	4.3	1.2
10	58.0	Male	0.8	0.2	298.0	33.0	59.0	6.2	3.1	1.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	63.0	Male	-11.618663736021448	6.193491797510766	379.0	277.0	80.0	5.829193415793082	2.908490151953772	0.7335331914063136
2	31.0	Female	0.6173859831704978	0.2675193935025096	782.0	65.0	147.0	6.408352034446754	3.3188771501833054	0.7812523026074208
3	21.0	Male	3.340277443487783	2.54733879544838	271.0	213.0	461.0	5.4241410447853	2.527804662103583	1.0760997473544249
4	24.0	Male	0.6636047516932884	0.17287969819593602	111.0	25.0	22.0	6.944261499848814	4.220473951189849	1.379752437246136
5	45.0	Female	0.8570172074424518	0.19162935566665665	353.0	20.0	-12.0	6.189415311891765	2.773148860592984	0.7683423771849744
6	31.0	Male	22.201207027893226	8.28132439968827	581.0	458.0	437.0	6.1840635163064	3.077664800913649	0.7961333187891824
7	44.0	Female	1.9	0.6	298.0	378.0	602.0	6.6	3.3	1.0
8	30.0	Male	0.8	0.2	182.0	46.0	57.0	7.8	4.3	1.2
9	53.0	Male	0.9	0.4	238.0	17.0	14.0	6.6	2.9	0.8
10	28.0	Male	0.8	0.3	190.0	20.0	14.0	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	44.0	Male	0.5805332522166781	0.11066152224770606	168.0	11.0	10.0	7.42360909241557	3.8292041783643285	1.1595982447217494
2	15.0	Male	0.9646992636157664	-0.03954437833575933	451.0	31.0	39.0	7.866645239606196	3.2878322813747727	1.0396912218215275
3	33.0	Male	1.1418799892569012	0.22933611895064676	171.0	27.0	35.0	6.273486402171462	4.648494009008318	0.9713732779309999
4	57.0	Male	14.199620004952628	2.5896980059498924	359.0	20.0	35.0	7.362067974038542	3.5100278553719355	0.7218156337584811
5	19.0	Male	-2.552028337364353	-0.27705299584567455	413.0	18.0	173.0	6.650316059798985	3.653167827233952	0.8972705925428219
6	59.0	Female	1.2739919963793178	0.27024395224632924	516.0	35.0	61.0	6.639744314464179	3.8612493641446632	0.9760895321771643
7	42.0	Male	0.7	0.2	197.0	64.0	33.0	5.8	2.4	0.7
8	69.0	Male	0.9	0.2	215.0	32.0	24.0	6.9	3.0	0.7
9	53.0	Male	1.6	0.9	178.0	44.0	59.0	6.5	3.9	1.5
10	49.0	Male	1.3	0.4	206.0	30.0	25.0	6.0	3.1	1.06

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65.0	Male	8.722652026562272	6.595361498548704	659.0	39.0	37.0	6.91315235670136	1.958617952624697	0.4994796892733674
2	17.0	Male	0.37531669938115547	0.09993311012180989	131.0	17.0	27.0	6.458993997205864	3.4978149111244763	1.340875792853739
3	41.0	Male	-4.469088672458291	7.384148381893995	654.0	488.0	338.0	6.997518400692448	2.767458555594466	1.989490925858727
4	57.0	Male	1.0118840020538662	0.3186202849580118	132.0	18.0	11.0	6.383321908851107	3.2301076956448975	1.115406020782096
5	60.0	Male	1.4742687439670936	0.4436040139573104	326.0	26.0	54.0	7.772210231090354	3.975005863814983	1.2
6	37.0	Male	8.451366523737656	2.847284169748951 5	312.0	28.0	84.0	6.109366243760247	1.815947621003467 7	0.2957841218188828 6
7	33.0	Male	2.1	0.7	205.0	50.0	38.0	6.8	3.0	0.7
8	22.0	Female	6.7	3.2	850.0	154.0	248.0	6.2	2.8	0.8
9	48.0	Male	5.8	2.5	802.0	133.0	88.0	6.0	2.8	0.8
10	84.0	Female	0.7	0.2	188.0	13.0	21.0	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65.0	Male	0.8108562887122805	0.4437344886256 92	46.0	62.0	64.0	6.269460229472119	2.527684910414215	0.7291354473800497
2	39.0	Female	1.5899279911884645	0.7750609766132 04	174.0	27.0	-35.0	5.208393159598183	2.777233889887666	1.013825662051266
3	16.0	Male	0.5356750787509253	0.5845556208891 89	321.0	72.0	86.0	7.113037378243449	4.4853513992646485	1.3832182724759303
4	59.0	Male	0.7495923307444357	0.3145524245262 56	274.0	82.0	146.0	6.958067449779596 5	2.577146036805115 3	0.7084372758620784
5	17.0	Male	2.400717579074686	0.4648345344907 9004	181.0	-7.0	52.0	6.091359787062787	2.392638746037253	1.3987950937871665
6	59.0	Male	0.8220279611596911	0.1623340128564 1222	225.0	6.0	38.0	8.525483729323627	4.211222751964 78	1.0276554152752126
7	50.0	Male	0.9	0.3	194.0	190.0	73.0	7.5	3.9	1.0
8	49.0	Male	0.6	0.1	218.0	50.0	53.0	5.0	2.4	0.9
9	18.0	Male	0.8	0.2	228.0	55.0	54.0	6.9	4.0	1.3
10	30.0	Female	0.7	0.2	194.0	32.0	36.0	7.5	3.6	0.92

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	24.0	Male	1.2867246921133204	0.1744261915385149	131.0	45.0	18.0	5.898939563818051	2.281283396437793	0.6731494564039987
2	25.0	Male	0.797326166344878	0.10542526825406182	458.0	36.0	50.0	6.828405754557335	3.802886260914023	1.241420374202804
3	63.0	Male	-8.64707614566217	7.65779354237178	206.0	324.0	437.0	8.309761719542152	2.695016714086793	1.4949198324985355
4	42.0	Male	0.5469477470976243	0.07320582904996462	251.0	39.0	43.0	6.017258872501989	2.969372641478075	0.6296453456453965
5	69.0	Female	3.366209533188513	-1.4864568465111336	-360.0	44.0	10.0	4.872941413799254	2.294799217390651	0.7252877222953874
6	52.0	Male	0.2909216588584277	0.2985697209482016	558.0	19.0	20.0	8.604129455106063	4.044363026119432	1.433214179774948
7	13.0	Female	0.7	0.2	350.0	17.0	24.0	7.4	4.0	1.1
8	54.0	Female	23.2	12.6	574.0	43.0	47.0	7.2	3.5	0.9
9	63.0	Male	0.9	0.2	194.0	52.0	45.0	6.0	3.9	1.85
10	46.0	Male	0.6	0.2	290.0	26.0	21.0	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	72.0	Male	1.2473086288819921	0.3738515400528414	93.0	32.0	132.0	6.931198107419063	3.4358523243143586	0.7413983855872444
2	44.0	Female	1.2351301377076582	0.08659338234753794	163.0	23.0	31.0	6.079624909873812	2.994847296854	1.0107529152783556
3	62.0	Male	1.0984632551549827	0.2966526397314324	215.0	28.0	20.0	6.773877345220566	2.547419812201604	0.7667143465128911
4	59.0	Female	0.7719104830590964	0.1864842193944236	251.0	26.0	48.0	6.468887774861897	3.280716090052296	0.7616202533347316
5	38.0	Male	3.3634931227537109	1.7142028668640299	353.0	58.0	27.0	4.346249876882396	1.7303894378128837	0.5568902076271194
6	20.0	Male	1.8287820696861243	0.4704410190346525	180.0	12.0	37.0	7.106616209204393	4.285571503128799	1.0256605801192786
7	48.0	Male	0.7	0.2	208.0	15.0	30.0	4.6	2.1	0.8
8	60.0	Male	22.8	12.6	962.0	53.0	41.0	6.9	3.3	0.9
9	27.0	Male	1.0	0.3	180.0	56.0	111.0	6.8	3.9	1.85
10	28.0	Female	0.8	0.2	309.0	55.0	23.0	6.8	4.1	1.51

TabuLLM: AI Benchmark Report

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	58.0	Male	0.79810188166143 6	0.17977079838570 437	791.0	67.0	62.0	4.18264180008904 3	2.90627590353064 76	0.72536996870308 77
2	52.0	Male	-9.12695229226188 3	9.09284155213635 6	451.0	447.0	601.0	7.07361867395004 7	3.17648236132462 23	0.67156226336511 93
3	70.0	Male	0.67778915257838 21	-0.15038150151195 864	129.0	19.0	27.0	5.07254303500542 3	2.22460797099817 63	0.92276207120386 52
4	40.0	Male	0.05046143315491 758	0.81814823357816 69	315.0	107.0	40.0	6.43639932539580 9	2.51501279417318 64	0.90956403383008 24
5	33.0	Female	2.63998119160778 04	0.04424936560540 038	393.0	119.0	396.0	6.837224507727	4.77899483292492 3	1.30907867748368 67
6	63.0	Male	0.87703330123167 97	0.36731766702362 39	190.0	14.0	21.0	5.01573933225832 9	3.64257036415765 1	1.00372481491906 67
7	26.0	Male	2.0	0.9	195.0	24.0	65.0	7.8	4.3	1.2
8	47.0	Male	2.7	1.3	275.0	123.0	73.0	6.2	3.3	1.1
9	74.0	Male	1.0	0.3	175.0	30.0	32.0	6.4	3.4	1.1
10	48.0	Male	3.2	1.6	257.0	33.0	116.0	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	39.0	Male	3.99361253484689 9	1.05773837178425 74	262.0	7.0	-56.0	8.78541950802629 8	4.20877428713190 1	1.39713223662794
2	36.0	Male	1.08184194923225 06	-0.43000152625727 717	482.0	113.0	-66.0	8.2326461895569	3.94020051451084 14	0.93381289297843 57
3	39.0	Male	0.78867402983445 86	0.15499780099639 102	367.0	42.0	102.0	5.32785879220707 15	1.83517312555688 64	0.48391224791728 277
4	23.0	Male	1.62003675662805 65	0.49205258993196 41	706.0	137.0	168.0	6.78452621907397 7	3.39290082305502 1	1.71673937091908 57
5	61.0	Male	28.5147448142050 3	10.8825002234462 54	515.0	37.0	174.0	5.82976016437547 1	2.09612879733468 25	0.51299436967235 56
6	43.0	Female	0.72044873993357 8	0.1	292.0	45.0	23.0	4.42945335505028 5	3.61301629205527 7	1.47271931000786 26
7	54.0	Female	5.5	3.2	350.0	67.0	42.0	7.0	3.2	0.8
8	43.0	Male	22.5	11.8	143.0	22.0	143.0	6.6	2.1	0.46
9	57.0	Male	4.5	2.3	315.0	120.0	105.0	7.0	4.0	1.3
10	65.0	Male	0.7	0.2	199.0	19.0	22.0	6.3	3.6	1.3